

Horizon Mobile Benchmark — Google AI Edge

Int4-Q4 vs Int8-Q8 · 60 Devices

June 2026

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Overview

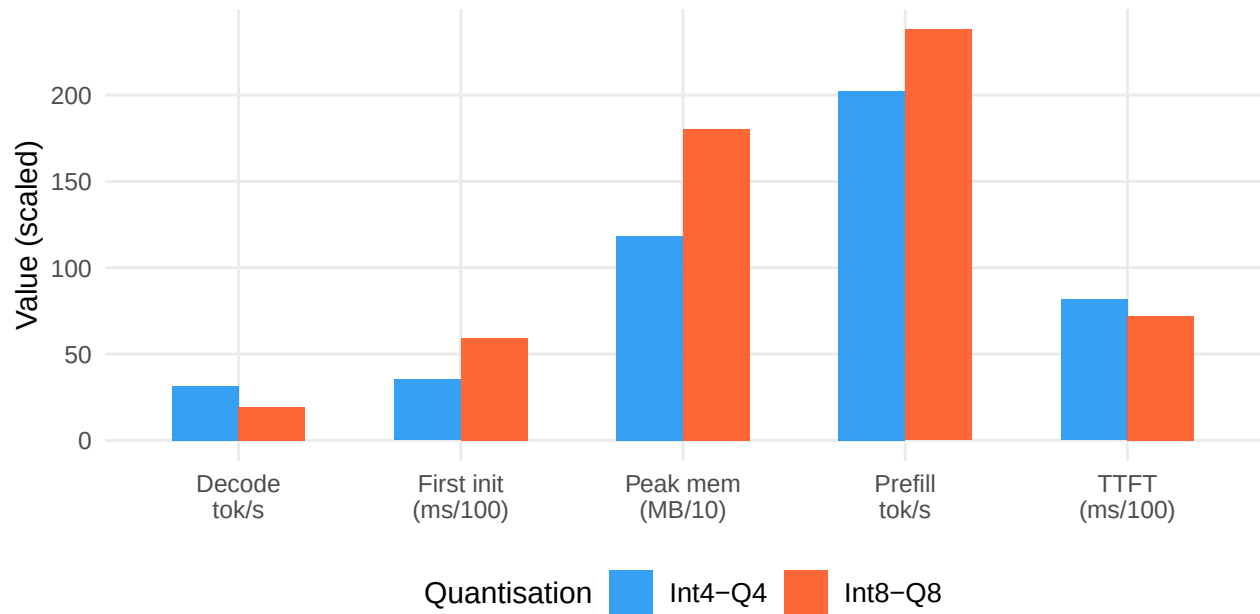
This report analyses **Horizon Mobile** benchmarked by Google AI Edge across **60 Android devices** (120 rows), comparing Int4-Q4 and Int8-Q8 quantisation.

Config	Tier	N	Decode (tok/s)	TTFT (ms)	Mem (MB)	Init (ms)
Int4-Q4	High tier	31	42.9	4377	1165	2027
Int4-Q4	Mid tier	22	21.2	9055	1150	3947
Int4-Q4	Low tier	7	12.7	22123	1376	8896
Int8-Q8	High tier	31	26.0	3674	1813	3129
Int8-Q8	Mid tier	22	13.7	8209	1799	7267
Int8-Q8	Low tier	7	8.0	19707	1785	14204

1. Q4 vs Q8 — All Metrics

Q4 vs Q8 – Global Average on Every Metric

Time/memory metrics scaled for comparability.

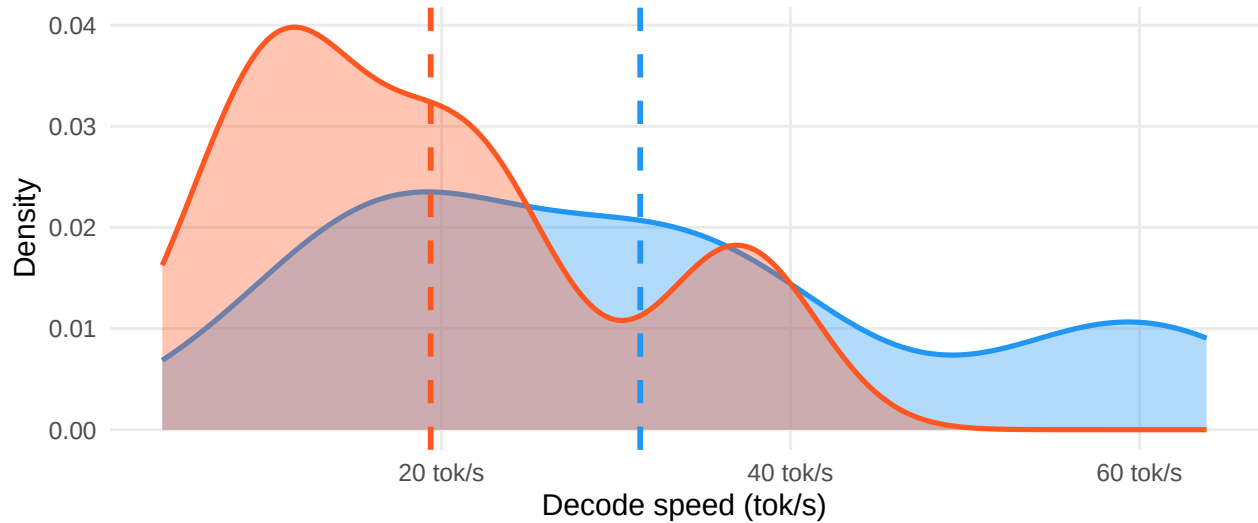


Metric	Q4 Avg	Q8 Avg	Q4/Q8	Winner
First init (ms)	3,532	5,938	0.60x	Q4
Steady init (ms)	2,612	3,470	0.75x	Q4
Time to first token (ms)	8,163	7,208	1.13x	Q8
Prefill (tok/s)	202	238	0.85x	Q8
Decode (tok/s)	31.4	19.4	1.62x	Q4
Peak memory (MB)	1,184	1,805	0.66x	Q4

2. Decode Speed Distribution

Decode Speed Distribution

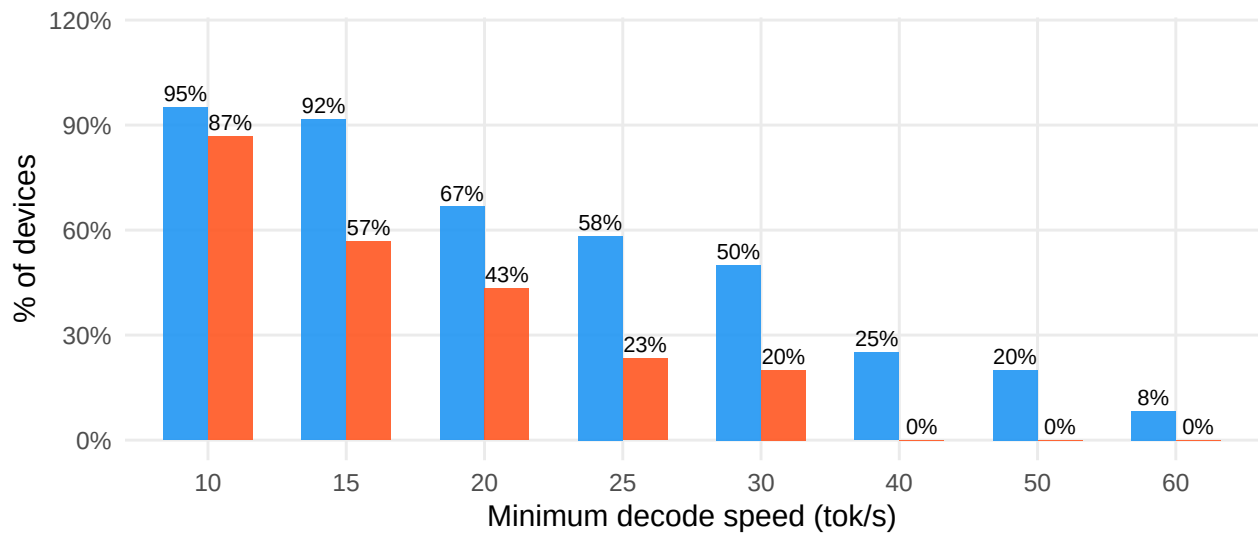
Dashed lines = group mean. Q4 is shifted right and wider.



Quantisation  Int4-Q4  Int8-Q8

% of Devices Meeting Decode Speed Threshold

No Q8 device reaches 40 tok/s. Q4 hits it on 25% of devices.

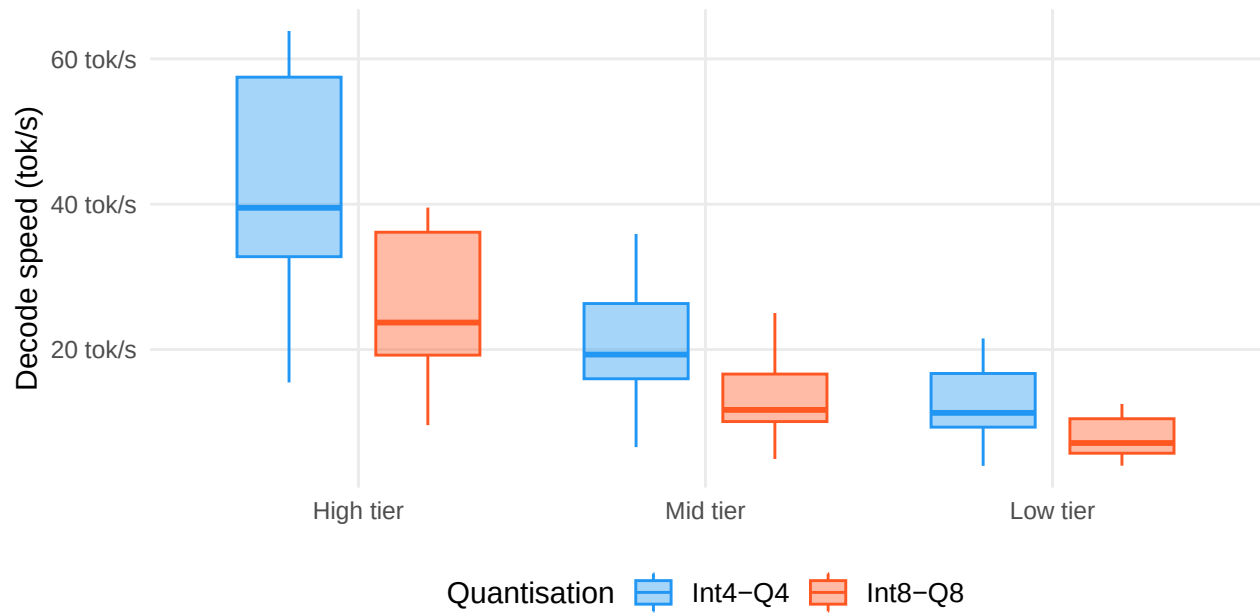


Quantisation  Int4-Q4  Int8-Q8

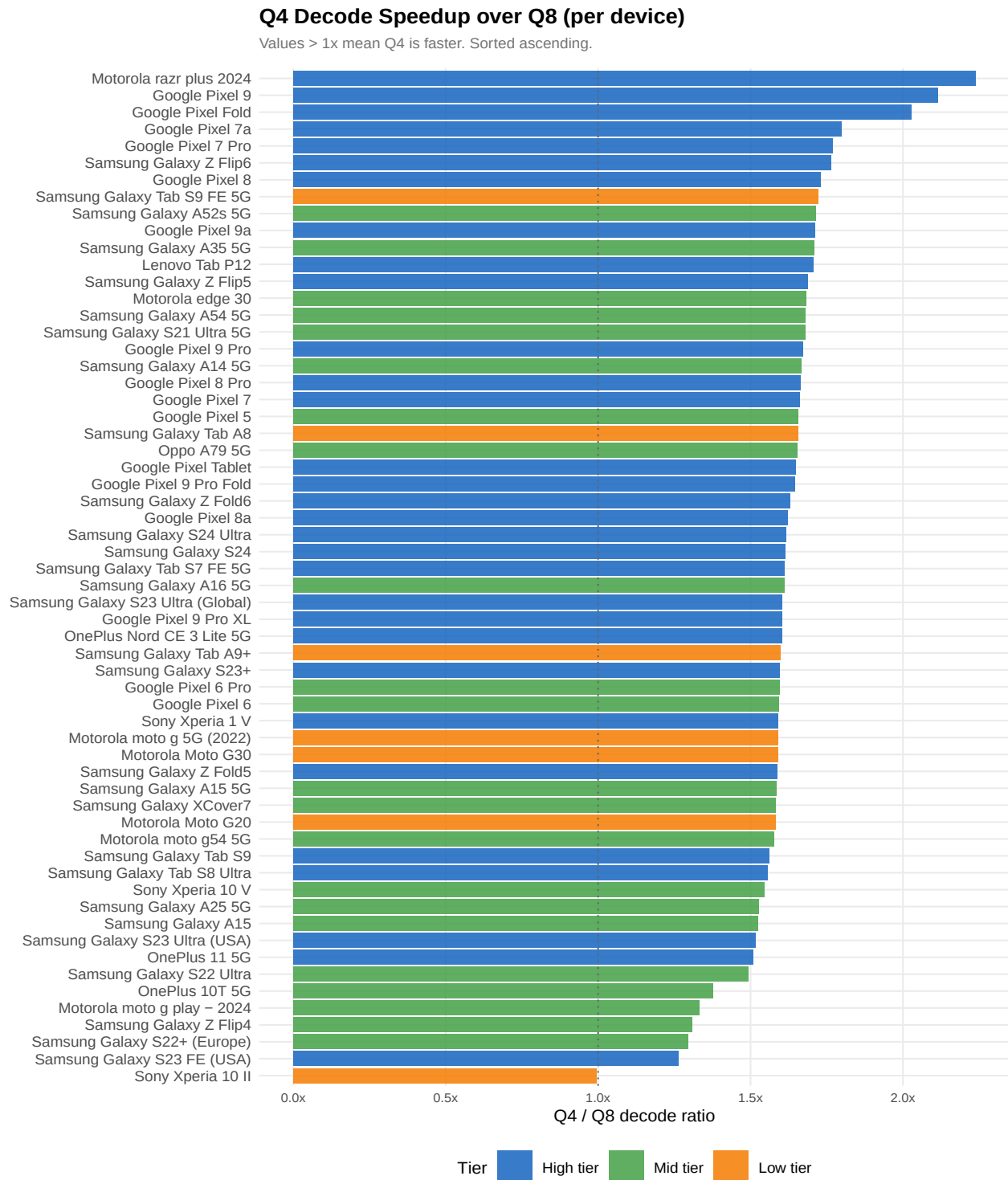
3. Decode Speed by Device Tier

Decode Speed by Device Tier

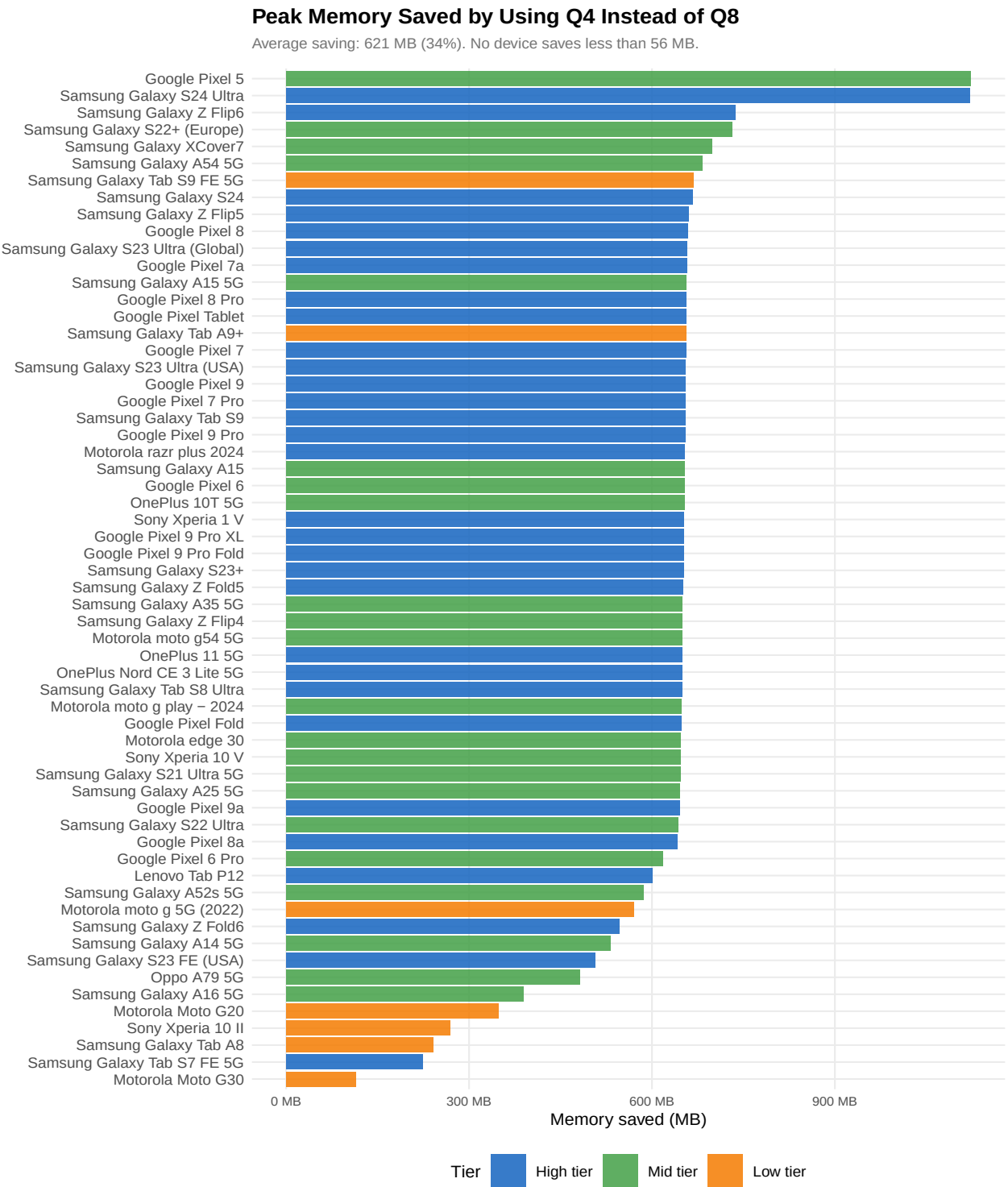
Q4 advantage is consistent across all tiers



4. Per-Device Q4 Decode Speedup Over Q8

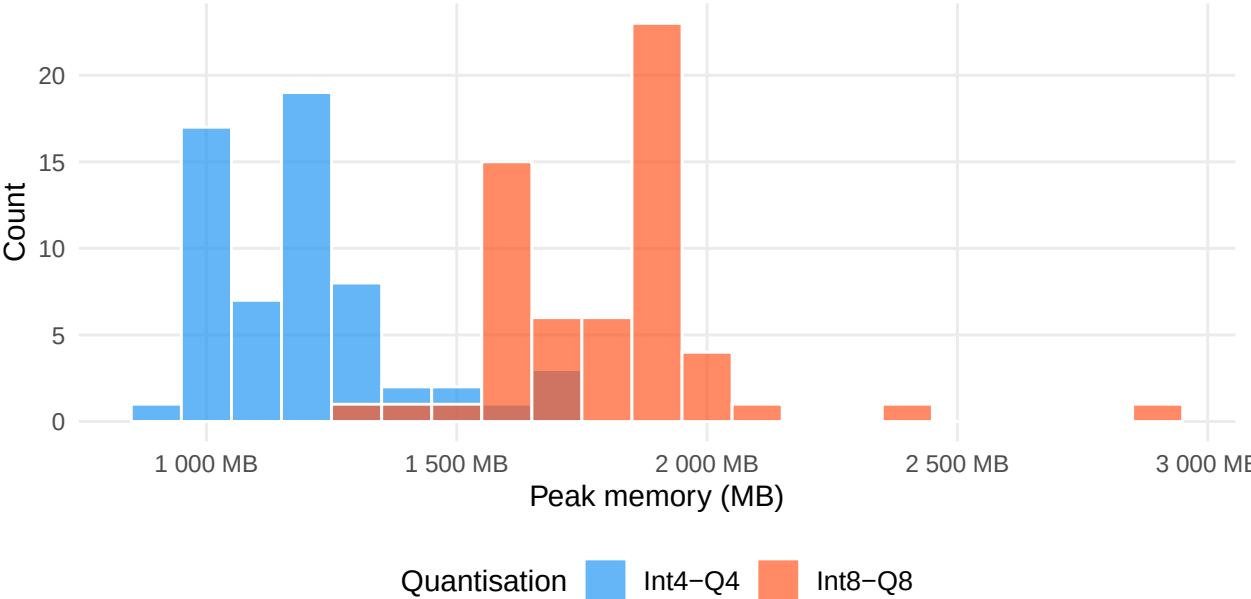


5. Memory Savings — Q4 vs Q8



Peak Memory Distribution

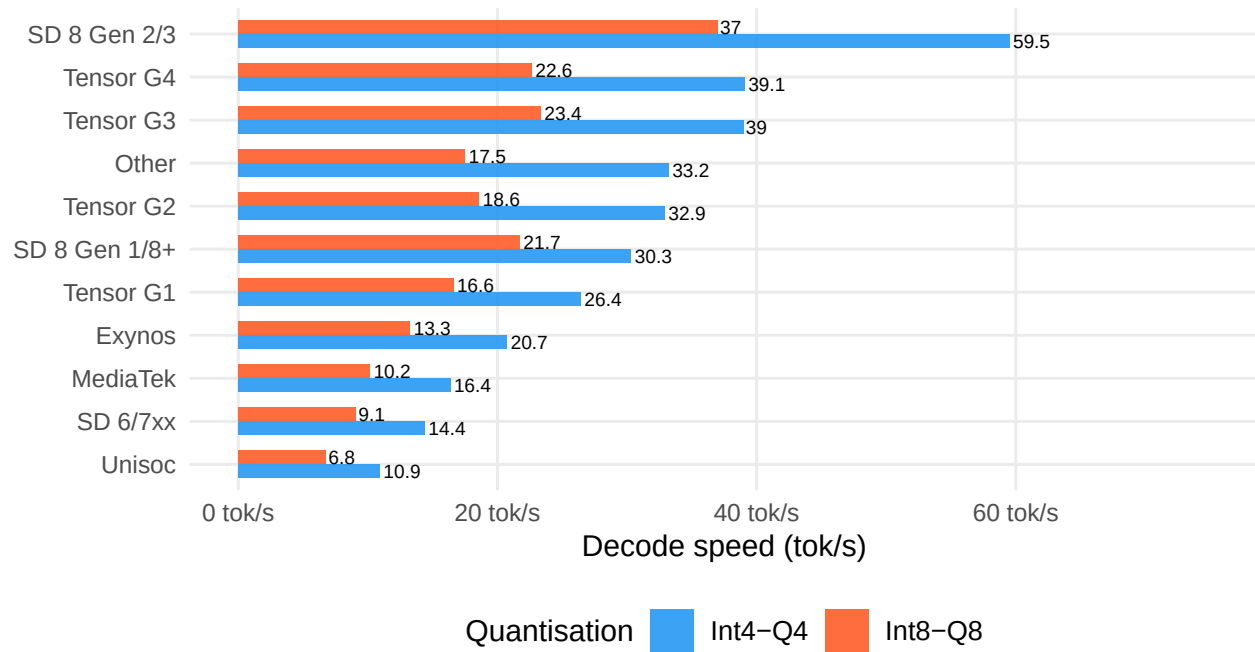
Q4 cluster: ~960–1,500 MB. Q8 cluster: ~1,600–2,450 MB.



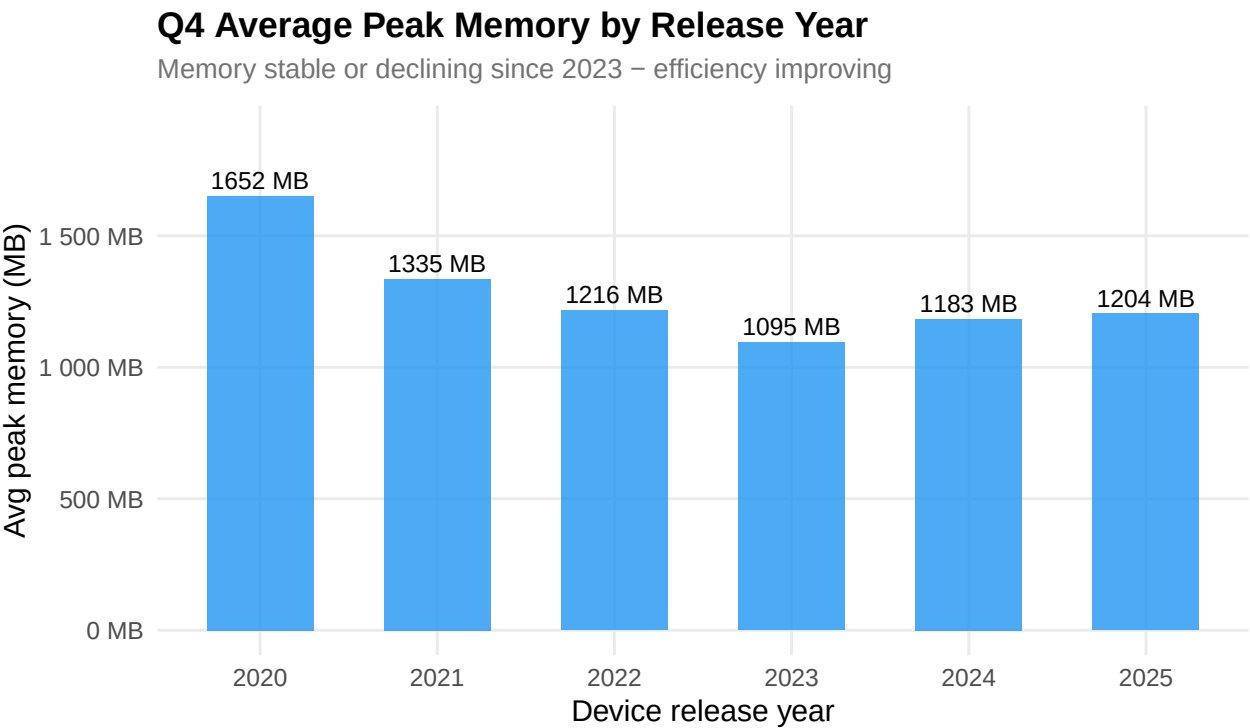
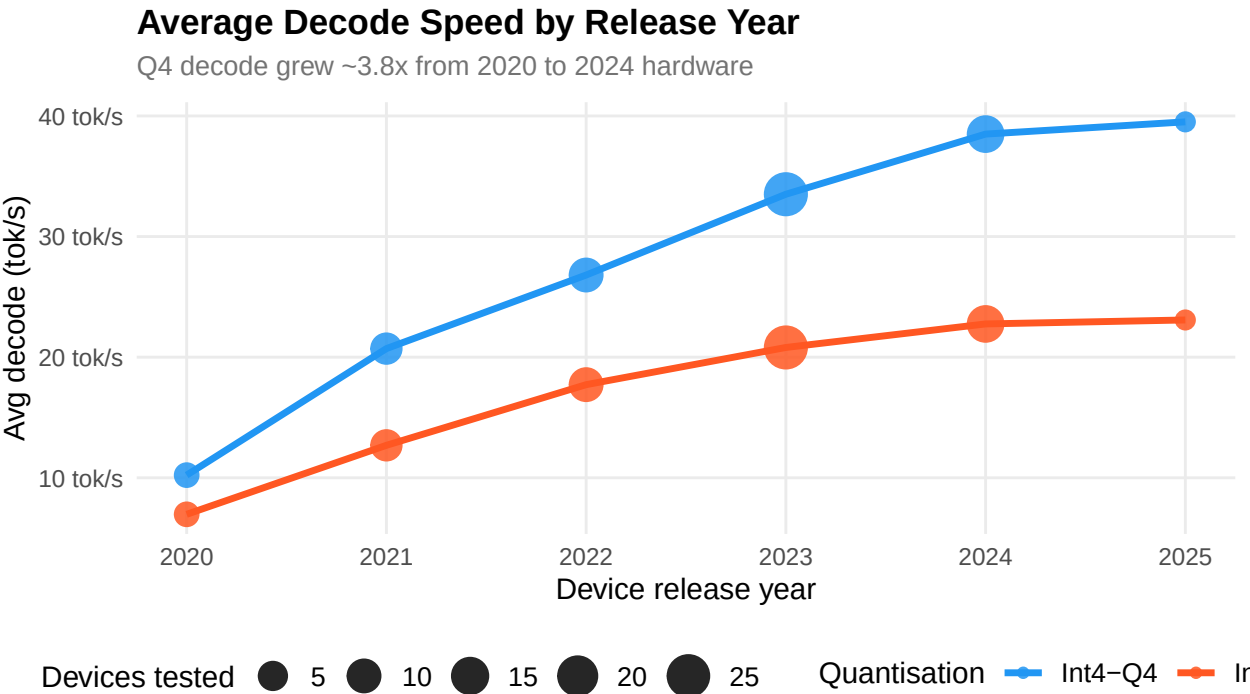
6. Processor Family Comparison

Average Decode Speed by Processor Family

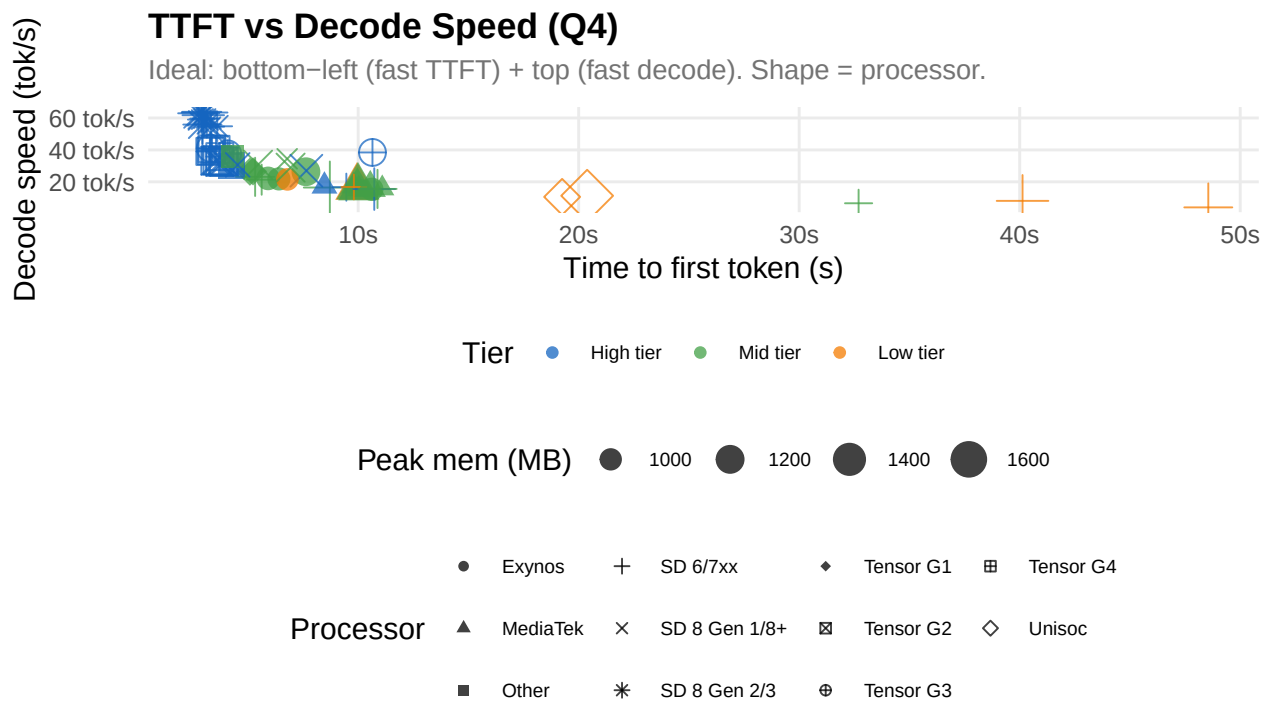
Snapdragon 8 Gen 2/3 dominates. Tensor G3/G4 lead ARM competitors.



7. Year-over-Year Progression



8. Time to First Token vs Decode Speed



9. Top 10 Devices — Q4 Decode

Device	Tier	Year	Processor	Decode	TTFT (ms)	Mem (MB)
Samsung S24	High tier	2024	Snapdragon 8 Gen 3	64	2907	1231
Samsung Z Fold6	High tier	2024	Snapdragon 8 Gen 3	63	3035	1464
Samsung Z Flip6	High tier	2024	Snapdragon 8 Gen 3	63	2768	1293
Samsung Z Flip5	High tier	2023	Snapdragon 8 Gen 2	63	2998	964
Samsung S24 Ultra	High tier	2024	Snapdragon 8 Gen 3	62	2975	1310
Samsung Tab S9	High tier	2023	Snapdragon 8 Gen 2	60	3102	970
Samsung S23+	High tier	2023	Snapdragon 8 Gen 2	60	2988	971
Samsung Z Fold5	High tier	2023	Snapdragon 8 Gen 2	59	2880	970
Samsung S23 Ultra (Global)	High tier	2023	Snapdragon 8 Gen 2	56	3138	970
Sony Xperia 1 V	High tier	2023	Snapdragon 8 Gen 2	56	2934	1204

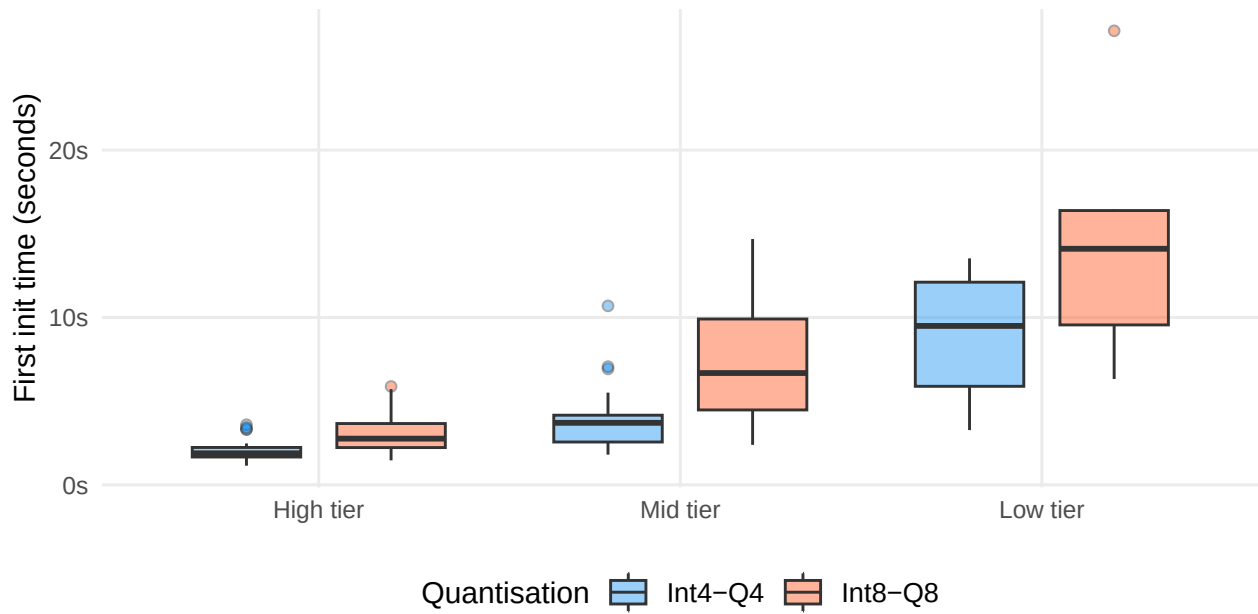
10. Worst 10 Devices — Q4 Decode

Device	Tier	Year	Processor	Decode	TTFT (ms)	Mem (MB)
Sony Xperia 10 II	Low tier	2020	Snapdragon 665	4	48553	1558
Motorola moto g play - 2024	Mid tier	2024	Snapdragon 680 4G	7	32694	978
Motorola Moto G30	Low tier	2021	Snapdragon 662	8	40129	1726
Samsung Galaxy Tab A8	Low tier	2022	Unisoc Tiger T618	11	19250	1142
Motorola Moto G20	Low tier	2021	Unisoc T700	11	20385	1736
Samsung Galaxy A14 5G	Mid tier	2023	MediaTek Dimensity 700	15	9962	1260
Samsung Galaxy A25 5G	Mid tier	2023	Exynos 1280	15	10620	1073
Samsung Galaxy Tab S7 FE 5G	High tier	2021	Snapdragon 750G 5G	15	10730	1315
Samsung Galaxy A15	Mid tier	2023	MediaTek Helio G99	16	10026	969
Sony Xperia 10 V	Mid tier	2023	Snapdragon 695 5G	16	10877	1206

11. Initialization Time

First Initialization Time by Tier

Q4 loads 40% faster on average. Outliers are legacy low-end chips.



Summary

Finding	Detail
Q4 decode is 1.62x faster	Consistent across all tiers and processor families
Q4 saves 34% memory	~621 MB saved on average; critical for 4 GB RAM devices
Q4 loads 40% faster	3.5s vs 5.9s cold start
Q8 wins on TTFT & prefill	12% faster first token; 18% faster prompt processing
No Q8 device breaks 40 tok/s	Q4 hits 40+ on 25% of tested devices
Snapdragon 8 Gen 2/3 leads	59.5 tok/s Q4 avg - 52% faster than next group
Tensor G3/G4 is competitive	~39 tok/s Q4, strong for a non-Qualcomm chip
Legacy devices are marginal	SD 662/665/680 and Unisoc below 10 tok/s
2020 to 2024 = 3.8x speed gain	Hardware progression is the biggest decode lever
Minimum viable: 2022+ mid-range	Anything older struggles to reach 20 tok/s